Review retrospective: *the second cumulant — susceptibility equals connected variance*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/AnharmonicSecondCumulant.lean`: the susceptibility (the h-derivative at 0 of the perturbed mean $\langle u\rangle_h = G_1(h)/G_0(h)$, $u = -(tx)$) equals the connected second cumulant $\langle u^2\rangle_0 - \langle u\rangle_0^2$ — the FDT-at-the-mean / κ_2 rung. Result: **a clean fidelity pass, no edits**.

1 Setting

The anharmonic arc computes the perturbed moments $G_n(h) = \int(-(tx))^n e^{-t(L+hx)}$. This file differentiates the mean G_1/G_0 at $h = 0$ (the susceptibility) and identifies it with the unperturbed variance of $u = -(tx)$, building on the moment-normalisation reviewed earlier this session ($\texttt{gibbsExp}((-(tx))^n) = G_n(0)/G_0(0)$).

2 What was reviewed

Two public theorems: `anharmonic_mean_hasDerivAt` (the quotient-rule derivative) and `anharmonic_susceptibility_eq_connected_variance` (its identification with $\langle u^2\rangle - \langle u\rangle^2$).

3 Fidelity / soundness findings

Sound; GPT found no mismatch.

- **Quotient rule**: with $f = G_1$, $g = G_0$ and the partition derivatives $f' = G_2$, $g' = G_1$ (each ∂_h brings a $-(tx)$ factor), `HasDerivAt.div` gives $\partial_h[G_1/G_0]|_0 = (G_2G_0 - G_1^2)/G_0^2$ — the correct $(f'g - fg')/g^2$.
- **Variance form**: substituting $\texttt{gibbsExp}((-(tx))^n) = G_n(0)/G_0(0)$ $(n = 1, 2)$ turns the RHS $\langle u^2\rangle - \langle u\rangle^2$ into $G_2/G_0 - (G_1/G_0)^2$, and the denominator-clearing algebra (`div_pow`/`div_sub_div`/`div_eq_div_iff`/`ring`, $G_0 \neq 0$) matches $(G_2G_0 - G_1^2)/G_0^2$.
- **Observable/sign**: the moments $(-(tx))^2$, $(-(tx))^1$ are exactly $\langle u^2\rangle$, $\langle u\rangle$ for $u = -(tx)$; the cumulant $\langle u^2\rangle - \langle u\rangle^2$ has no sign slip.

4 Simplifications applied

None. GPT's (B) findings are all golf/refactor (inline `hc`/`hd`; a `calc` for the four rewrites; `field_simp [hne]; ring` for the final algebra; a shared `hne` helper) — declined (golf not reliablysafe; helper-extraction out of single-file scope). No dead import/open, and all four hypotheses are consumed via the cited derivative/positivity lemmas.

5 Disagreements and open questions

None.

6 Lessons

- **The κ_2/susceptibility review is the quotient rule plus the moment-normalisation bridge.** Two checks carry it: the quotient-rule numerator $(G_2G_0 - G_1^2)$ (needing $\partial_h G_n = G_{n+1}$), and the algebraic identity $(G_2G_0 - G_1^2)/G_0^2 = G_2/G_0 - (G_1/G_0)^2$ that ties the quotient form to the variance form via the previously-reviewed moment-normalisation. Both are mechanical once the cumulant ladder's partition-derivative and normalisation lemmas are trusted.

7 Follow-ups

- The κ_3 rung (`AnharmonicThirdCumulant`) and the cumulants-from-log-partition file are the natural next reviews in this arc.